Alex Leung

Senior SWE • Tech Lead • Experimenter • Lifelong Student

University of Washington 2014 - 2018
Bachelor of Science in Computer Science
with an Entrepreneurship minor

at Flexport
in Bellevue
from Aug 2022
to Mar 2026

Tech Lead / Staff Software Engineer

Foundational Services
`TypeScript, GraphQL, Node, Next, Envoy, Terraform, K8s, OAuth2, RBAC, AWS`

- Championed operational excellence across GraphQL stack. Reduced all API request latency by 50%. Improved backend reliability from below 90% to over 99% by architecting fine-grained error attribution system and driving company-wide accountability.
- Standardized authorization across company, both GraphQL authz and service-to-service authz
- Got cross company consensus on new dynamic service discovery standard using AWS CloudMap. This merged several competing standards at the company.
- Consolidated ingress across all apps / external APIs into a single entrypoint for improved security / simplicity; led team of 4 to create new unified API gateway
- Drove consolidation of similar libraries and developer tools across multiple codebases. Measured DevX improvements quantitatively using developer velocity metrics, code reduction, and developer surveys and qualitatively using developer anecdotes and interviews
- Led Flow to TypeScript migration for legacy websites and tools
- Led migration from Algolia to OpenSearch for one division, saving $500k per year
- Architected microfrontends framework for Flexport. Lead team of 7 engineers to create this; onboarded six separate orgs, become foundation for Flexport's most profitable applications

at Hulu
Remotely
from Sep 2021
to Feb 2022

Tech Lead / Senior Software Engineer

Flex `TypeScript, GraphQL, Node.js, Express, Spinnaker`

- Drove adoption of our cross-platform UI-as-data microservice "Flex", it allows simple, generic device rendering code and rapid experimentation by configuration
- Architected migration of Flex from a RESTful API to GraphQL subgraph for the new company-wide federated GraphQL service created as part of the Hulu, Disney+ unification effort
- Forged partnership w/ similar Disney team. We took over UI as data API, they i18n as a service

at Amazon
in Seattle
from Sep 2017
to Sep 2021

Software Development Engineer II

Fire TV `Java, Kotlin, Docker, AWS, DynamoDB`

Designed, implemented microservice responsible for ingestion of IMDB movies and shows catalog

Last Mile Identity `Java, AWS, Cognito, Coral (RPC), Lambda`

- Extended OAuth microservice to support AuthN of various user types
- Maintained role-based AuthZ (RBAC) microservice and guided system redesign

Last Mile Account Management `Java, AWS, Coral (RPC), Elasticsearch, React`

- Designed and implemented microservices as part of legacy monolith rearchitecture
- Created hybrid web-app views in React

Last Mile Risk and Compliance `Java, AWS, Angular, mTLS, Coral (RPC), SQL`

- Integrated background check microservice with external vendor REST APIs
- Microservices for in-app facial recognition of package deliverers (en-route ID check)
- Created data warehousing pipelines from AWS Dynamo DB tables to AWS Redshift
- Automated code deployment (CI/CD pipeline building)

Payments Experience `Scala, Java, jQuery, SCSS, Coral (RPC)`

- Designed and implemented one-time-password verification for payment screens
- Integrated with away teams' authentication services

at Leasera
in Seattle
from Aug 2016
to Sep 2017

Lead Software Engineer

`TypeScript, GraphQL, React, React Native, Ruby on Rails, Jenkins, AWS, MySQL`

- Built Ruby on Rails ERB / jQuery views then later migrated these to React + GraphQL
- Designed MySQL schema. Optimized DB migrations, SQL transactions, ORM code
- Maintained server infrastructure; configured EC2, Jenkins, Nginx, MySQL, Redis
- Led project to create new mobile app using React Native, Redux, and Apollo GraphQL
- Integrated 3rd-party APIs such as Stripe, Zillow, TransUnion, Twilio, Mailgun, etc.
- Lead team in goal planning and managing delivery timelines. Reported sprint progress to CEO

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TECHNICAL SKILLS

`Maintainability: DRY, Design Patterns, SOLID, Service Oriented Architecture, Majestic Monoliths, Result Error Pattern`

`Front-End: Expo, Next.js, Remix, React, React Native, Vue, Ionic, Angular, jQuery, HTML, SCSS, Tailwind, Webpack, Web Workers, Tamagui`

`Server-Side: Node.js, Kafka, PostgreSQL, SQLite, Neo4j, Cockroach DB, Redis, DynamoDB, Serverless (ex. AWS Lambda), Convex, SpacetimeDB`

`Languages: TypeScript, Effect TS, Java, Kotlin, Scala, Rust, Ruby, Solidity, Golang, Python, C#, C, C++, Ruby, PHP, SQL, HCL`

`Dev Infrastructure: AWS, GCP, Github Actions, Buildkite, ArgoCD, K8s, IaC, CDK, Terraform, Docker, Linux, Envoy, Nginx, Apache, Spinnaker, Backstage`

`Protocols: gRPC, GraphQL, REST, TLS`

`Crypto / AuthZN: OAuth2, Asymmetric Key Crypto, Digital Signatures, mTLS, RBAC, pRBAC`

`Web 3: EVM, Solana, Move`

`Observability: Open Telemetry, Statsig, Datadog, Grafana, Loki, Prometheus`